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Three-dimensional simulation of heat and moisture transfer in woven fabric structures

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Abstract

Fabric structure parameters have a significant impact on the comfort of heat and moisture transfer in garments. Previous numerical simulations required extensive mathematical calculations and mostly investigated one- or two-dimensional models of fiber assembly without considering the weave structure, which is a key parameter, that significantly influences the porosity of the woven structure and consequently its heat and moisture management. While the finite element method supports the simulation of the optimal shape and material properties with better visibility, previous finite element models focused on heat transfer and neglected water vapor transfer in fabrics. In this article, the finite element simulation of heat and moisture transfer in woven fabrics is established based on the testing principle of thermal resistance and moisture resistance tester using COMSOL Multiphysics software. In this simulation, three-dimensional parametric geometrical models of the fabric are created using curve interpolation methods by acquiring the control point coordinates of different weaves (plain, 2/2 balanced twill, and 4/1 unbalanced twill weaves). Heat and moisture transfer properties of fabric models in the horizontal and vertical directions were analyzed, including the heat flux, moisture resistance, water vapor permeability, and water vapor concentration. The article also deals with the effects of weave structure and fabric cover in a range of 72.1–85.1% on the fabric heat flux and water vapor concentration. Comparison between model and experimental results revealed that the three-dimensional simulation can accurately predict the impact of weave pattern and fabric cover on the fabric heat and moisture transfer performance. In addition, this model can be utilized to study the distribution of heat and water vapor within fabrics, providing a theoretical foundation for optimizing heat and moisture comfort in woven fabrics.

Keywords

Three-dimensional modeling, finite element simulation, heat and moisture transfer, weave structure, woven fabric, structural parameters

The heat and moisture transfer of clothing is crucial for the comfort of end users. When temperatures rise, the body sweats, and gaseous and liquid water accumulate between the skin and clothing. Poor moisture permeability of clothing causes a sticky feeling and negatively impacts thermal sensation.[1]

Realizing the importance of heat and moisture management of clothing and their relation to end users' comfort, there have been extensive studies on the thermal properties and humidity of fabrics during the past few decades. Scholars have used numerical analysis to acquire analytical solutions for heat, humidity, and their dynamic variations.[2] As early as 1948, Henry[3] established a mathematical model to analyze the heat and moisture transfer properties of fiber assembly. The following two equations were derived by applying the mass balance and conservation of energy principle, respectively:

$$\varepsilon \frac{\partial C_a}{\partial t} + (1-\varepsilon)\frac{\partial C_f}{\partial t} = \frac{D_a \varepsilon}{\tau} \frac{\partial^2 C_a}{\partial x^2}$$

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$$C_v \frac{\partial T}{\partial t} - (1-\varepsilon)\lambda \frac{\partial C_f}{\partial t} = K \frac{\partial^2 T}{\partial x^2}$$

where ε is the porosity, C_a $(kg \cdot m^{-3})$ is the concentration of water vapor in the inter-fiber pores, C_f $(kg \cdot m^{-3})$ is the concentration of water vapor in the fibers, D_a $(m^2 \cdot s^{-1})$ is the liquid diffusion coefficient, x (mm) is the location along the thickness of fabric, T $(^\circ C)$ is the temperature, t (s) is the time, C_v $(J \cdot m^{-3} \cdot ^\circ C)$ is the volumetric heat capacity of the fabrics, λ $(J \cdot kg^{-1})$ is the heat of adsorption, and K $(W \cdot m^{-1} \cdot ^\circ C^{-1})$ is the coefficient of thermal conductivity.

The study by Henry ignored the adsorption kinetics of fibers and properties of the material itself, and could not explain the process of heat and moisture transfer well as the moisture and heat transfer are functions of fiber type. Today, clothes are manufactured from numerous commercially available natural and synthetic fibers with a broad range of fiber properties.

In 2010, Zhu et al.[4] developed a fractal mathematical model to investigate the impact of porosity on water vapor transfer and the temperature change in plain weave fabrics, incorporating sorption/desorption and evaporation/condensation mechanisms. In 2013, Xu et al.[5] proposed a dynamic model of fiber assembly under low temperature, considering the thickness, sorption and condensation during heat and moisture transfer. In 2015, Fontana et al.[6] presented a prediction model for an insulating coated fabric, with the addition of a mechanism of heat radiation energy transfer. In 2022, Yang et al.[7] established a model to predict the cooling performance of knitted interlock fabrics made from hydrophobic fibers, while taking into account the rate of airflow.

The models above primarily analyzed the flow within the fiber layer in one or two dimensions under atmospheric pressure conditions and ignored the three-dimensional (3D) nature of textile fabrics. In addition, these methods differ from the actual scenarios that typically necessitate a significant number of mathematical analyses and calculations, which were not taken into consideration. Conversely, the application of the finite element (FE) method, which was not considered, can define and couple a range of partial differential equations, while considering the transfer properties of the material. This enhances the accuracy and efficiency of predictive models and allows for visualizing the numerical results to determine the optimal shape and material properties.

In 2020, Zhang et al.[8] simulated the heat transfer properties of fabrics with different structural parameters (yarn spacing, yarn crimp height, and yarn cross-section width and thickness) and the model considered the effective thermal conductivity of yarns and the anisotropy of fibers. In 2022, Codau et al.[9] simulated the variation of temperature and heat flux of plain weave fabrics under steady-state and dynamic-state conditions. Liu et al.[10] utilized the FE model to study the thermal conductivities of yarn along the yarn axial and transverse directions of three high-temperature-resistant woven fabrics made from inorganic fibers at different temperatures, considering yarn density and the width and thickness of yarn cross-section. Numerous scholars have examined the heat transfer process in fabrics at various levels, including different layers, high temperatures, and heat leakage.[11–15] While these models resulted in improved prediction over the earlier models, they focused on heat transfer and neglected water vapor transfer in fabrics.

In 2020, Yang et al.[16] simulated weft-knitted fabrics to study the effect of different yarn fineness on their heat and moisture transfer performance. In 2022, Gu et al.[17] utilized a FE model to investigate the effect of different fiber orientation and porosity on the heat and moisture transfer to the polyurethane fiber membrane on a microscopical basis. Nonetheless, these models failed to investigate the effects of different fabric structures (such as weave type, yarns density, yarn spacing, and yarn diameter) on the heat and moisture transfer of the fabric.

An essential mechanism for the transfer of water vapor in fabrics occurs through their opening vents (interstices between fibers and yarns), which rely on fabric porosity. Effective parameters that influence the porosity of a fabric are weft and warp yarn density, warp and weft yarn twist, warp and weft yarn type (or spinning method), thread count, and weave pattern.[18] Numerical simulations and experiments have demonstrated that differences in the shape and arrangement of pores in porous media, etc. can cause fluctuations in their effective thermal conductivity and moisture phase change.[19–22] In addition, the fabric cover has a significant effect on the moisture permeability, air permeability, and thermal resistance of fabrics.

Woven fabrics are widely used in apparel and their moisture and heat management influence the comfort of the end users. Therefore, it is crucial to investigate how the fabric's structural parameters affect heat and moisture transfer. In this study, the heat and moisture transfer in woven fabrics with different weave pattern cover and fabric cover are analyzed by establishing 3D simulation utilizing FE method using COMSOL Multiphysics software. The 3D simulation predicts the distribution of heat and water vapor inside woven fabrics. Predicted results driven from the simulation are validated experimentally, demonstrating that the 3D model can provide a tool for designing woven fabrics with optimal structural parameters to achieve heat and

moisture comfort that potentially lead to the reduction/elimination of costly experimental trials.

3D simulation methodology

The 3D simulation and experimental validation includes several sequential steps; these are:

1. *Fabric geometrical modeling*: The fabric's structural parameters are determined based on its interweaving characteristics and geometric features. The bending control point of the yarn axis is derived, and the yarn is numerically characterized. Afterwards, the modeling software's sweep, array, and other tools are used to create a 3D geometric model of the fabric.
2. *Construction of the FE model*: Based on the testing principle of SGHP thermal resistance and wet resistance tester, the selection of FE physical field and the setting of boundary conditions are carried out, and the parameters required for FE simulation are input.
3. *Verification of the validity of FE model*: Three equal-support woven fabric samples constructed from the same yarn (polyester filament yarn, 900D) with different weave structures (plain, 2/2 balanced twill, and 4/1 unbalanced twill weaves) were chosen for experimentation to test their moisture resistance and water vapor permeability. This is because the plain weave and 2/2 twill belong to balanced weave fabric, but their weave factor is different. And the 4/1 twill belongs to unbalanced weave fabric. Therefore, these three kinds of fabrics have the characteristics of different fabric structure, which are representative to a certain extent, and can be used to study the effect of different fabric structure and organization on heat and moisture transfer. The actual structural parameters and material properties of the fabric samples were input into the FE software to create a 3D model. The software carries out numerical solution after mesh division, leading to simulated values of moisture resistance and water vapor permeability. These simulated values were compared to the experimental values measured using SGHP tester to validate the FE model's accuracy.

Geometry modeling of fabrics

Determination of coordinates of yarn control points. It was assumed that the warp and weft yarns are incompressible uniform cylinders and same yarn number (diameter), and that the warp and weft yarns spacing are identical (square fabrics). The geometry of the plain weave fabric that fits the assumptions is shown in Figure 1.

In Figure 1, L_j denotes the adjacent warp yarn spacing and L_w denotes the adjacent weft yarn spacing, under the assumption that the yarn spacing is the same, $L_j = L_w = L$; H_j denotes the wave height of the warp yarn, H_w denotes the wave (crimp) height of the weft yarn, and D is the yarn diameter, $H_j = H_w = D$, and the thickness of the fabric $\delta = 2D$.

The spatial trajectory of the yarn in the woven fabric can be determined according to the geometric relationship shown above, so as to establish a 3D geometric model. Taking plain weave fabric as an example, according to the interweaving law, it is known that 5 control points are needed to determine an axis of plain weave structure, including three equilibrium position points and two optimum points of flexion wave.[23,24] Similarly, it is known that the 2/2 twill weave fabric structure of an axis needs 9 control points, and the 4/1 twill weave fabric structure of an axis needs 11 control points. Figure 2(a)–(c) display the positions of the control points on axis of warp1 in the plain weave fabric structure, the 2/2 twill fabric structure, and the 4/1 twill fabric structure, respectively. According to the interweaving law, it is possible to determine the coordinates of the control points of the warp and weft yarns' axes within one organizational cycle of the three fabrics sequentially, as listed in Tables 1–3.

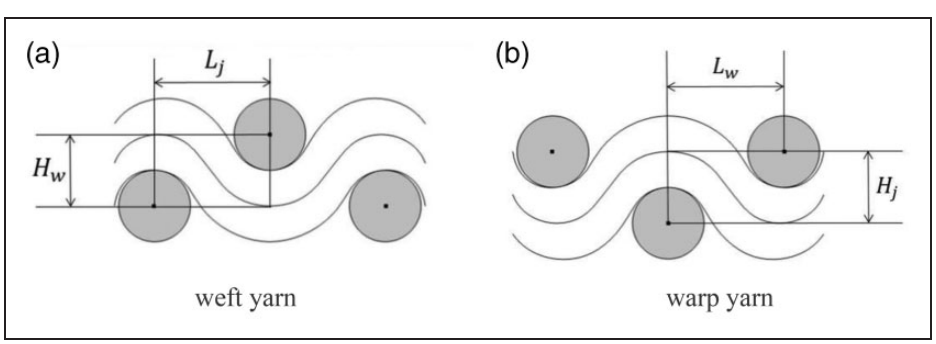

Figure 1. Geometry of plain weave fabric: (a) weft yarn and (b) warp yarn.

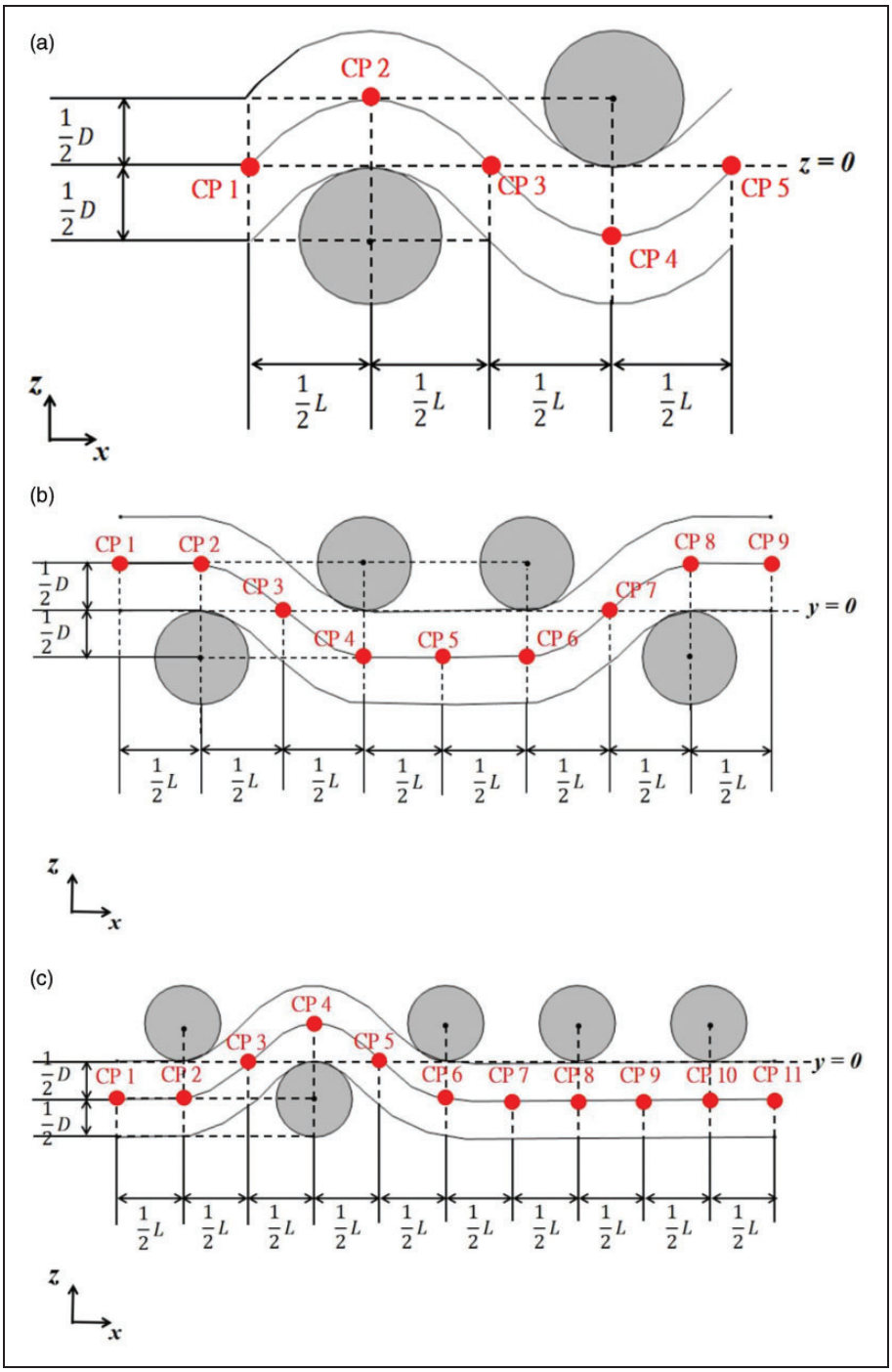

Figure 2. Position of control points (CPs) on axis of warp 1 in fabric structures: (a) plain weave fabric, (b) 2/2 twill weave fabric, and (c) 4/1 twill weave fabric.

Table 1. Plain weave fabric yarn axis curve control point (CP) coordinates

Yarn	CP 1	CP 2	CP 3	CP 4	CP 5
Warp1	[0,0,0]	[L/2,0,D/2]	[L,0,0]	[3L/2,0,–D/2]	[2L,0,0]
Warp2	[0,–L,0]	[L/2,–L,–D/2]	[L,L,0]	[3L/2,–L,D/2]	[2L,–L,0]
Weft1	[L/2,L/2,0]	[L/2,0,–D/2]	[L/2,–L/2,0]	[L/2,–L,D/2]	[L/2,–3L/2,0]
Weft2	[3L/2,L/2,0]	[3L/2,0,D/2]	[3L/2,–L/2,0]	[3L/2,–L,–D/2]	[3L/2,–3L/2,0]

Table 2. 2/2 Balanced twill weave fabric yarn axis curve control point (CP) coordinates

Yarn	CP 1	CP 2	CP 3	CP 4	CP 5	CP 6	CP 7	CP 8	CP 9
Warp1	[0,0,D/2]	[L/2,0,D/2]	[L,0,0]	[3L/2,0,–D/2]	[2L,0,–D/2]	[5L/2,0,–D/2]	[3L,0,0]	[7L/2,0,D/2]	[4L,0,D/2]
Warp2	[0,–L,0]	[L/2,–L,D/2]	[L,–L,D/2]	[3L/2,–L,D/2]	[2L,–L,0]	[5L/2,–L,–D/2]	[3L,–L,–D/2]	[7L/2,–L,–D/2]	[4L,–L,0]
Warp3	[0,–2L,–D/2]	[L/2,–2L,–D/2]	[L,–2L,0]	[3L/2,–2L,D/2]	[2L,–2L,D/2]	[5L/2,–2L,D/2]	[3L,–2L,0]	[7L/2,–2L,–D/2]	[4L,–2L,–D/2]
Warp4	[0,–3L,0]	[L/2,–3L,–D/2]	[L,–3L,–D/2]	[3L/2,–3L,–D/2]	[2L,–3L,0]	[5L/2,–3L,D/2]	[3L,–3L,D/2]	[7L/2,–3L,D/2]	[4L,–3L,0]
Weft1	[L/2,L/2,0]	[L/2,0,–D/2]	[L/2,–L/2,–D/2]	[L/2,–L,–D/2]	[L/2,–3L/2,0]	[L/2,–2L,D/2]	[L/2,–5L/2,D/2]	[L/2,–3L,D/2]	[L,–7L/2,0]
Weft2	[3L/2,L/2,D/2]	[3L/2,0,D/2]	[3L/2,–L/2,0]	[3L/2,–L,–D/2]	[3L/2,–3L/2,–D/2]	[3L/2,–2L,–D/2]	[3L/2,–5L/2,0]	[3L/2,–3L,D/2]	[3L/2,–7L/2,D/2]
Weft3	[5L/2,L/2,0]	[5L/2,0,D/2]	[5L/2,–L/2,D/2]	[5L/2,–L,D/2]	[5L/2,–3L/2,0]	[5L/2,–2L,–D/2]	[5L/2,–5L/2,–D/2]	[5L/2,–3L,–D/2]	[5L/2,–7L/2,0]
Weft4	[7L/2,L/2,–D/2]	[7L/2,0,–D/2]	[7L/2,–L/2,0]	[7L/2,–L,D/2]	[7L/2,–3L/2,D/2]	[7L/2,–2L,D/2]	[7L/2,–5L/2,0]	[7L/2,–3L,–D/2]	[7L/2,–7L/2,–D/2]

Table 3. 4/1 Unbalanced twill weave fabric yarn axis curve control point (CP) coordinates

Yarn	CP 1	CP 2	CP 3	CP 4	CP 5	CP 6	CP 7	CP 8	CP 9	CP 10	CP 11
Warp1	[0,0,–D/2]	[L/2,0,–D/2]	[L,0,0]	[3L/2,0,D/2]	[2L,0,0]	[5L/2,0,–D/2]	[3L,0,–D/2]	[7L/2,0,–D/2]	[4L,0,–D/2]	[9L/2,0,–D/2]	[5L,0,–D/2]
Warp2	[0,–L,–D/2]	[L/2,–L,–D/2]	[L,–L,–D/2]	[3L/2,–L,–D/2]	[2L,–L,0]	[5L/2,–L,D/2]	[3L,–L,0]	[7L/2,–L,–D/2]	[4L,–L,–D/2]	[9L/2,–L,–D/2]	[5L,–L,–D/2]
Warp3	[0,–2L,–D/2]	[L/2,–2L,–D/2]	[L,–2L,–D/2]	[3L/2,–2L,D/2]	[2L,–2L,–D/2]	[5L/2,–2L,–D/2]	[3L,–2L,0]	[7L/2,–2L,D/2]	[4L,–2L,0]	[9L/2,–2L,–D/2]	[5L,–2L,–D/2]
Warp4	[0,–3L,0]	[L/2,–3L,–D/2]	[L,–3L,–D/2]	[3L/2,–3L,–D/2]	[2L,–3L,–D/2]	[5L/2,–3L,–D/2]	[3L,–3L,–D/2]	[7L/2,–3L,–D/2]	[4L,–3L,0]	[9L/2,–3L,D/2]	[5L,–3L,0]
Warp5	[0,–4L,0]	[L/2,–4L,D/2]	[L,–4L,0]	[3L/2,–4L,–D/2]	[2L,–4L,–D/2]	[5L/2,–4L,–D/2]	[3L,–4L,–D/2]	[7L/2,–4L,–D/2]	[4L,–4L,–D/2]	[9L/2,–4L,–D/2]	[5L,–4L,0]
Weft1	[L/2,L/2,0]	[L/2,0,D/2]	[L/2,–L/2,D/2]	[L/2,–L,D/2]	[L/2,–3L/2,D/2]	[L/2,–2L,D/2]	[L/2,–5L/2,D/2]	[L/2,–3L,D/2]	[L/2,–7L/2,0]	[L/2,–4L,–D/2]	[L/2,–9L/2,0]
Weft2	[3L/2,L/2,0]	[3L/2,0,–D/2]	[3L/2,–L/2,0]	[3L/2,–L,D/2]	[3L/2,–3L/2,D/2]	[3L/2,–2L,D/2]	[3L/2,–5L/2,D/2]	[3L/2,–3L,D/2]	[3L/2,–7L/2,D/2]	[3L/2,–4L,D/2]	[3L/2,–9L/2,0]
Weft3	[5L/2,L/2,D/2]	[5L/2,0,D/2]	[5L/2,–L/2,0]	[5L/2,–L,–D/2]	[5L/2,–3L/2,0]	[5L/2,–2L,D/2]	[5L/2,–5L/2,D/2]	[5L/2,–3L,D/2]	[5L/2,–7L/2,D/2]	[5L/2,–4L,D/2]	[5L/2,–9L/2,D/2]
Weft4	[7L/2,L/2,D/2]	[7L/2,0,D/2]	[7L/2,–L/2,D/2]	[7L/2,–L,D/2]	[7L/2,–3L/2,0]	[7L/2,–2L,–D/2]	[7L/2,–5L/2,0]	[7L/2,–3L,D/2]	[7L/2,–7L/2,D/2]	[7L/2,–4L,D/2]	[7L/2,–9L/2,D/2]
Weft5	[9L/2,L/2,D/2]	[9L/2,0,D/2]	[9L/2,–L/2,D/2]	[9L/2,–L,D/2]	[9L/2,–3L/2,D/2]	[9L/2,–2L,D/2]	[9L/2,–5L/2,0]	[9L/2,–3L,–D/2]	[9L/2,–7L/2,0]	[9L/2,–4L,D/2]	[9L/2,–9L/2,D/2]

Establishing the 3D fabric geometrical models. The utilization of COMSOL's own geometric model construction system can effectively avoid the problem of incompatibility that may arise during the importation of models from other software. We used the following process. Insertion of the interpolation function in COMSOL software, input of the coordinates of each control point, and construction of the curve, as shown in Figure 3(a). Then in the x–y plane, respectively, we took the control point 1 of each yarn as the circle center, drawing the circle with the diameter D representing the cross-section of the yarn, as shown in Figure 3(b). We used the sweep tool to choose the circle as the surface to be swept, and interpolation curves as the edges to be followed, to construct the object, which could obtain a minimum fabric cycle of the plain structure, as shown in Figure 3(c). The full model of plain tissue could be achieved after array repetition, as shown in Figure 3(d). An air domain was created around the yarn to form the model of the fabric system containing stationary air, forming the union of yarn and air, as shown in Figure 3(e). To prevent

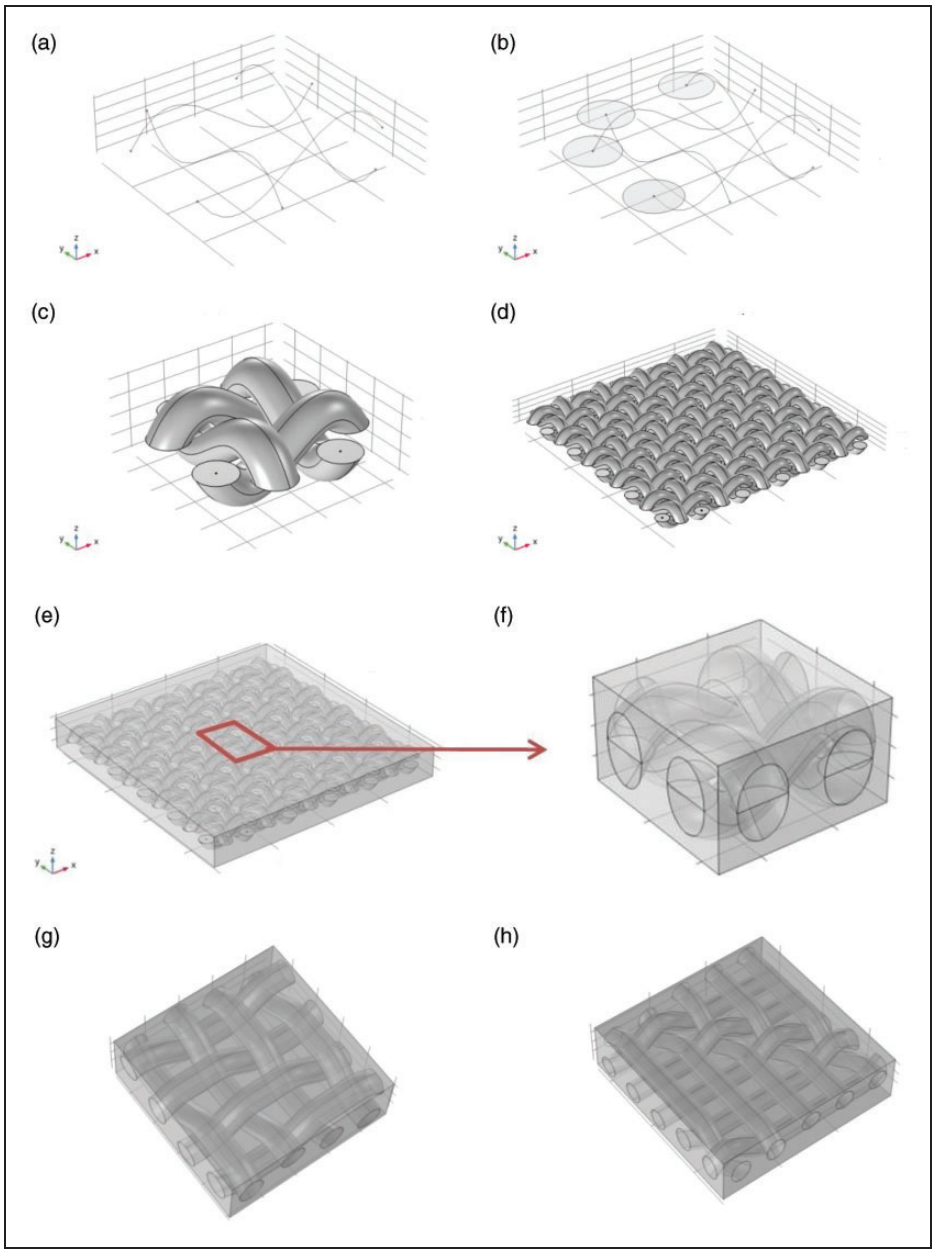

Figure 3. Fabric model construction process: (a) interpolation curve, (b) sweep curve, (c) plain weave fabric repeat, (d) multiple repeats of plain weave fabric, (e) the union of fabric and air, (f) final plain weave fabric repeat model, (g) final 2/2 twill weave fabric model, and (h) final 4/1 twill weave fabric model.

uneven meshing and reduce computation time, we thickened the air domain at the tangent point of the yarn by 0.02 mm. This change was minimal and had almost no effect on the simulation results. Lastly, the final fabric model was obtained by intercepting a tissue loop at the center, as shown in Figure 3(f). For the finite element simulation, utilizing the fabric model of one tissue cycle was sufficient, which could representatively reflect the heat and moisture transfer performance of the whole fabric, and could avoid the problem of complicated calculation caused by too large a model. Similarly, the final fabric model of 2/2 twill and 4/1 twill weave fabrics could be obtained, as shown in Figure 3(g) and Figure 3(h).

Construction of the FE model

Test principles. The physical field setup for the FE model had to be defined according to specific test conditions, which entailed testing for moisture resistance and water vapor permeability values. According to the ISO-11092-Ret test standard, a thermal resistance moisture resistance tester (SGHP) was used to determine the moisture resistance (R_{et}, Pa·m^2·W^{-1}) and water vapor permeability (W_d, g·m^{-2}·h^{-1}·Pa^{-1}) of the woven samples. The schematic diagram of the moisture resistance tester is presented in Figure 4. During the experiment, the test panel's surface temperature was set at 35°C/308.15 K, which is equivalent to the temperature of human skin. The ambient temperature was also set to 35°C/308.15 K; the glass fiber filter covered the top of the test panel, making it air-permeable and water-impermeable. As a result, the fabric's bottom surface was humidified with 100% saturated water vapor, while the ambient humidity was set at 40%. Due to the difference in humidity, water vapor will move from high concentration area to low concentration area, and during this process heat is transferred. To maintain a constant temperature of 35°C, heat flow was provided to the test board. The heat flow provided should be equal to the heat flow taken away during the transfer of water vapor, so that the fabric can be calibrated for moisture resistance. This method is commonly known as "skin model" and can replicate the heat and moisture transfer process that occurs close to the human skin. According to ISO-11092-Ret test standard, R_{et} is defined as the ratio of the difference in air pressure on both sides of the fabric to the evaporative heat flux passing vertically through the unit area of the sample. It is expressed as follows:

$$R_{et} = \frac{(P_{skin} - P_{amb})}{\frac{Q}{A}} \tag{1}$$

$$P_{skin} = 133.3 \cdot 10^{[8.10765-(1750.29/(235+T_{skin}))]} \tag{2}$$

$$P_{amb} = RH \cdot 133.3 \cdot 10^{[8.10765-(1750.29/(235+T_{amb}))]} \tag{3}$$

where P_{skin} (Pa) is the air pressure on the surface of the test plate, T_{skin} (°C) is the temperature on the surface of the test plate, P_{amb} is the ambient air pressure (Pa), T_{amb} (°C) is the temperature of the test environment, Q (W) is the heating power provided to the test plate, A (m^2) is the test area, and RH (%) is the ambient relative humidity.

According to the ISO-11092-Ret standard, water vapor permeability W_d is defined as the quantity of water vapor passing vertically through a unit area of the sample per unit time at a unit water vapor pressure. It can be calculated from:

$$W_d = \frac{1}{R_{et} \times \varphi_t} \tag{4}$$

where φ_t is the latent heat of evaporation of saturated water vapor at the test plate surface temperature T_{skin}. When $T_{skin} = 35°C$, then $\varphi_t = 0.627$ W·h·g^{-1}.

FE model setup and boundary conditions. Taking plain weave fabric as an example, the boundary condition of the fabric model is shown in Figure 5. According to the testing principle, the whole process of water vapor transfer is designed for fluid flow, heat transfer, and moisture transfer. Therefore, in the heat transfer

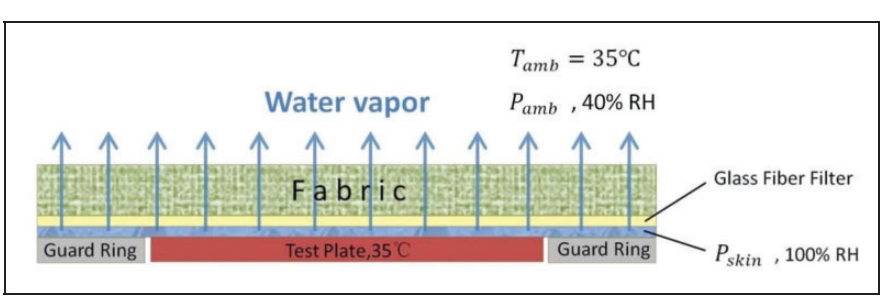

Figure 4. Schematic diagram of thermal resistance and moisture resistance tester (SGHP).

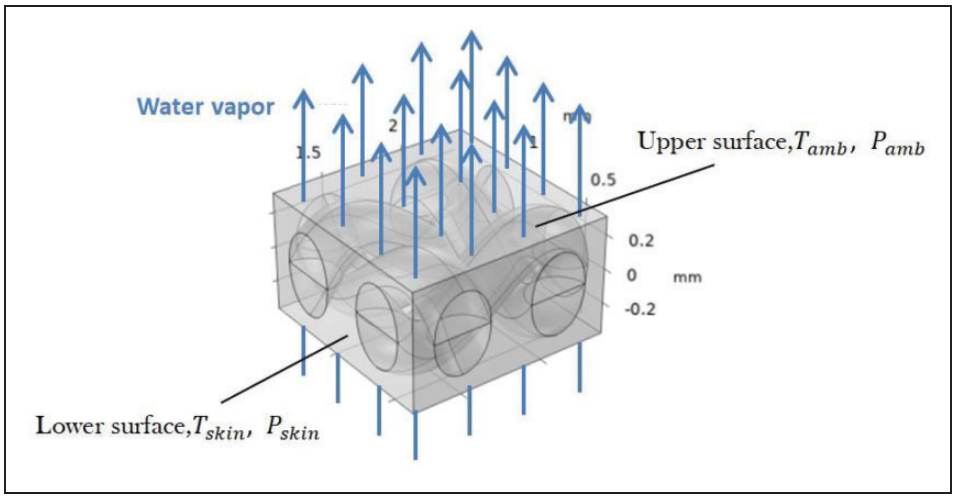

Figure 5. Boundary delineation of the plain weave fabric three-dimensional (3D) model.

module, solid heat transfer was selected, and the lower surface of the model was set as the entrance boundary with the temperature of 308.15 K, the upper surface was set as the open boundary with the temperature of 308.15 K, and the surrounding surfaces were thermally insulated, with an initial temperature of 308.15 K. Dilute substance transfer was selected in the chemical substance transfer module, this is because even at the simulated temperature of 308.15 K, the molar coefficient of saturated water vapor is only 5% of the molar coefficient of air, so the wet air can be regarded as a kind of dilute substance suitable for dilute substance transfer. The lower surface was set as the entrance boundary, that is, the water evaporation side, and the water vapor concentration was 2.194 mol/m^3, the upper surface was set as the open boundary, and the external concentration was 40% RH, that is, 0.8776 mol/m^3, and there was no flux on the surrounding surfaces, and the initial value was also 0.8776 mol/m^3. The same settings were applied to the 2/2 twill weave fabrics and 4/1 twill weave fabrics.

The yarn region was defined as a porous medium, with solid material properties based on the yarn material. Required parameters included yarn density ρ (kg/m^3), thermal conductivity k (W·m^{-1}·K^{-1}), and constant pressure heat capacity Cp (J·kg^{-1}·K^{-1}); the peripheral space was established as a fluid domain, specifically, an ideal gas with a density of 1.184 kg/m^3, thermal conductivity of 0.026 W·m^{-1}·K^{-1}, and constant pressure heat capacity of 1003.62 J·kg^{-1}·K^{-1}.

Governing equations for FE simulation. Based on the testing principle, the experimental process involves both heat and moisture transfer. When the temperature gradient is minimal, we can ignore heat radiation. The model can be considered as a combination of wet air and fabric, where heat transfer encompasses fluid heat transfer and solid phase heat transfer. Primarily, heat transfer in solids depends on heat conduction and satisfies Fourier's law:[25]

$$\vec{q} = -k\nabla T \tag{5}$$

For the transient problem, the solid heat transfer equation[25] is:

$$\rho Cp\frac{\partial T}{\partial t} = \nabla \cdot (k\nabla T) + Q \tag{6}$$

where ρ (kg/m^3) is the density of the solid, Cp is the constant pressure heat capacity of the solid (J/kg^{-1}·K^{-1}), T (°C/K) is the temperature, t (s) is the time, $\vec{q}$ (W/m^2) is the thermal conductivity flux of the solid, Q (W/m^3) is the heat of the solid, and k (W·m^{-1}·K^{-1}) is the thermal conductivity of the solid.

The fluid heat transfer is mainly thermal convection and follows the governing equations:[26]

$$\rho' Cp'\frac{\partial T}{\partial t} + \rho' Cp'\vec{u} \cdot \nabla T + \nabla\vec{q}' = Q' \tag{7}$$

$$\vec{q}' = -k'\nabla T \tag{8}$$

where ρ' (kg/m^3) is the fluid density, Cp' (J·kg^{-1}·K^{-1}) is the constant pressure heat capacity of the fluid, $\vec{u}$ (m/s) is the thermal convection velocity of the fluid, $\vec{q}'$ (W/m^2) is the thermal conductivity of the fluid, Q' (W/m^3) is the fluid's heat, k' (W·m^{-1}·K^{-1}) is the thermal conductivity of the fluid.

On wet transfer, the molar coefficient of saturated water vapor is also only 5% of the molar coefficient of air at the simulated temperature condition of 308.15 K.

Hence, the wet air can be considered as a rarefied substance, and Fick's law is fulfilled when water vapor (the solute) diffuses in air (the solvent) during the process:[27]

$$\vec{Ji} = -Di\nabla ci \tag{9}$$

During the simulation of the model, one side of the fabric is fully saturated with water vapor while the other side has a humidity level of 40% The water vapor concentration varies between the two sides, resulting in diffusion from higher to lower concentration. Under transient conditions, wet air transfer follows the control equation:[28]

$$\frac{\partial ci}{\partial t} + \vec{u} \cdot \nabla ci + \nabla \vec{Ji} = Ri \tag{10}$$

where ci (mol/m^3) is the water vapor concentration, $\vec{Ji}$ (mol·m^{-2}·s^{-1}) is the diffusion flux of water vapor, $\vec{u}$ (m/s) is the convective velocity of the gas phase, Di (m^2/s) is the diffusion coefficient of the gas phase, Ri (m^3·Pa·mol^{-1}·K^{-1}) is the ideal gas constant.

The transfer process follows the continuity and momentum conservation equations:[16]

$$\rho\nabla\vec{u} = 0 \tag{11}$$

$$0 = -p\nabla\vec{I} + \mu\nabla\cdot[\nabla\vec{u}+(\nabla\vec{u})^T]+F \tag{12}$$

where $\vec{u}$ (m/s) is the Darcy or apparent flow rate, p (Pa) is the pressure, ρ (kg/m^3) is the density of the material, μ (Pa·s) is the dynamic viscosity, F (N) is the mass force, and T denotes the transpose of the matrix.

Verification of the effectiveness of the FE model

To ensure the validity of the finite element model, samples of plain, 2/2 twill, and 4/1 twill weave fabrics made from the same material were chosen. Polyester (PET, full name of Polyethylene Terephthalate, a most used synthetic fiber) filament yarn was used as the raw material for the fabric samples due to its excellent hydrophobic properties. We tested the fundamental properties of the experimental fabric samples, including moisture resistance and water vapor permeability. Then, we input the fabric samples' parameters into the FE model, solved the model to obtain a set of simulated values, and compared these values with the corresponding test results to analyze and verify the efficacy of the FE model.

Acquisition of experimental data. The thickness of the experimental fabric samples was measured using the YG(B) 141D digital fabric thickness meter. The warp and weft density of the fabric were measured using a density mirror. The fabric porosity (ε) can be obtained based on Archimedes' principle: cut the fabric sample into $10 \times 10\,\text{cm}^2$ pieces (V_f). Prepare three specimens of each sample fabric and find the average value to reduce errors. Use a clean measuring cup and filled with 50 ml distilled water. Slowly submerge the fabric sample cloth into the measuring cup until completely submerged. After standing for 2 h in stable laboratory conditions, observe the rise in liquid level in the measuring cup. The increased volume represents the fabric yarn solid volume (V_y), excluding the pores. Using the following equation, we can obtain the porosity (ε) of the fabric:

$$\varepsilon = 1 - \frac{V_y}{V_f} \times 100\% \tag{13}$$

where ε is the fabric porosity, V_y (cm^3) is the fabric yarn solid volume, V_f (cm^3) is the total volume of the fabric sample cloth.

The weave factor (M) expresses the amount of yarn interlacing due to weave. It may be used as a measure of fabric tightness to compare fabrics if all other construction parameters of the fabrics are the same.[27] The weave factors were calculated from

$$M_{1(2)} = \frac{N_{2(1)}}{i_{1(2)}} \tag{14}$$

where N_1 and N_2 are the number of warp and filling yarns per weave repeat, respectively; i_1 and i_2 are the number of filling and warp intersections per weave repeat, respectively.

The thermal resistance and wetness resistance tester were utilized to determine the wetness resistance (R_{et}, Pa·m^2·W^{-1}) and water vapor permeability (W_d, g·m^{-2}·W^{-1}·Pa^{-1}) of the samples according to the ISO-11092-Ret test standards. Table 4 presents the various experimental data obtained from the fabric samples.

Acquisition of simulation results. Fabrics can be categorized based on the type of support surface as warp-supported fabrics, weft-supported fabrics, and equal-supported fabrics. The equal-supported fabric is a type of fabric with a warp and weft density ratio close to 1.[29] The experimental fabrics described in Table 4 are considered equal-supported fabrics. The interweaving law of the fabric follows the fifth structural phase, which suggests that the yarn spacing $L = \sqrt{3}D$.[30] These fabrics are known as square fabrics in which warp and weft geometric parameters are identical. Consequently, the fabric's 3D geometric model can be created using the geometric modeling method in the previous section.

Table 4. Parameters of fabric samples from polyester fiber

Weave type	Weave factor M	Fabric thickness δ (mm)	Warp density (ends/10 cm)	Weft density (picks/10 cm)	Yarn diameter $D = \delta/2$ (mm)	Fabric weight (g/m$^{2)}$	Porosity (%)	R_{et} (Pa. m^2·W^{-1})	W_d (g. m^{-2}·W^{-1}·Pa^{-1})
Plain	1.0	0.68	173	169	0.340	340	55.26	2.51	0.64
2/2 Twill	2.0	0.70	174	172	0.350	345	60.34	2.41	0.66
4/1 Twill	2.5	0.71	176	174	0.355	343	67.23	2.20	0.73

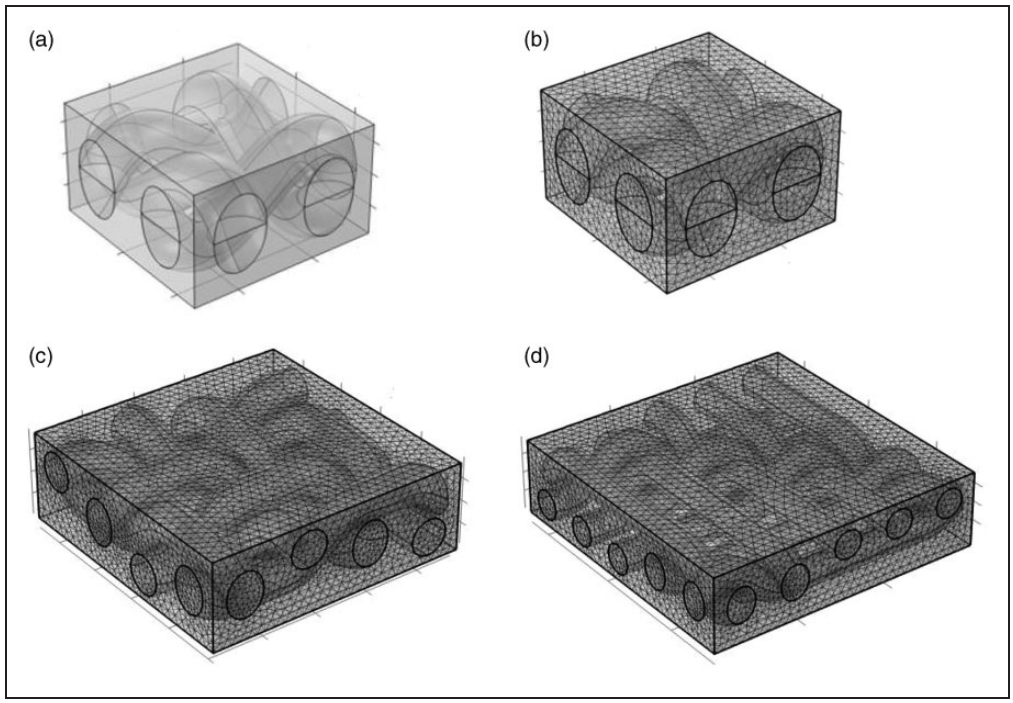

Figure 6. Geometric modeling and meshing: (a) three-dimensional (3D) model of plain weave fabric, (b) meshed plain weave fabric, (c) meshed 2/2 twill weave fabric, and (d) meshed 4/1 twill weave fabric.

For instance, a plain weave fabric's geometric model is depicted in Figure 6(a).

The fabric model was imported into the FE analysis COMSOL software. Overlapped and fused parts were removed via Boolean operation before meshing was done with a free tetrahedral mesh. Cell size was chosen to be conventional, and after automatic meshing, the complete mesh contained 97,782 domain cells, 10,552 boundary cells, and 947 edge cells with a maximum cell size of 6.97×10^{-5} m, and a minimum cell size of 2.08×10^{-5} m, and average mesh quality was 0.6615. A meshed plain weave fabric model is depicted in Figure 6(b). The procedures were applied to the 2/2 twill and 4/1 twill square fabrics to obtain their meshed structures shown in Figure 6(c) and Figure 6(d), respectively.

The material properties of the fabric model were defined based on polyester fiber material, where the density is 1380 kg/m^3, the constant pressure heat capacity is 1340 J·kg^{-1}·K^{-1}, and thermal conductivity is 0.084 W·m^{-1}·K^{-1}. The boundary conditions were established following the method in the previous section "Construction of the FE model."

Since the yarns were treated as a porous medium, the percentage of pores between the fibers in the yarns (yarns' pores) were considered as input to the model as the porosity of the medium. The porosity for the whole fabric system was considered, which includes the fluid domain excluding the yarn and the yarns' pores. The volume function in the FE can be used to acquire the fluid domain volume (V_a), and then it can be found as a percentage (ε_a) of the entire fabric model volume (V_c). Next, by subtracting ε_a from the total porosity ε, the porosity of the porous medium ε_b could be determined. The following equations demonstrate this process:

$$\varepsilon_a = \frac{V_a}{V_c} \tag{15}$$

$$\varepsilon_b = \varepsilon - \varepsilon_a \tag{16}$$

The calculation of porosity of yarns for plain, 2/2 twill, and 4/1 twill weave fabrics were 0.319%, 0.322%, and 0.327%, respectively.

The mathematical model was solved by constructing the FE model using COMSOL software. The solution was obtained through iterative calculations employing the FE method to solve the partial differential equations, facilitated by the steady state solver.

Comparison of experimental and simulated values. Based on the moisture resistance calculation equations (1)–(3), and the moisture permeability calculation from equation (4), the heat flux and water vapor permeability through the model in the FE simulation were compared with those converted from the experimentally measured values. The results of the comparison are presented in Figure 7, where the FE simulation outcomes are compared to the experimental data for three different fabrics. The error bar range in Figure 7(a) and (b) is only 3%.

The results indicate that the heat flux and water vapor permeability errors obtained through simulations aligned with the experimental values are all within 5%. Notably, the water vapor permeability simulation error of the plain weave fabric is only 2.05%. Although the simulated values for heat flux and water vapor permeability are slightly higher than the actual values, this discrepancy may have resulted from porosity settings. In the context of porosity, the measured porosity accounts for the fabric's total porosity, including closed pores caused by yarn slippage, pores on the outer surface, and the yarn hairiness protruding out of the fabric surface, which would result in a higher simulated porosity. The margin of error falls within acceptable limits, suggesting that the method has practical value and is capable of accurately modeling the heat and moisture transfer in fabrics. In addition, the actual yarns are not uniform cylinders as assumed for establishing the geometric model and the experimental measurement of the fabric thickness may not represent the fabric thickness mounted in thermal resistance and moisture resistance tester (SGHP) during testing.

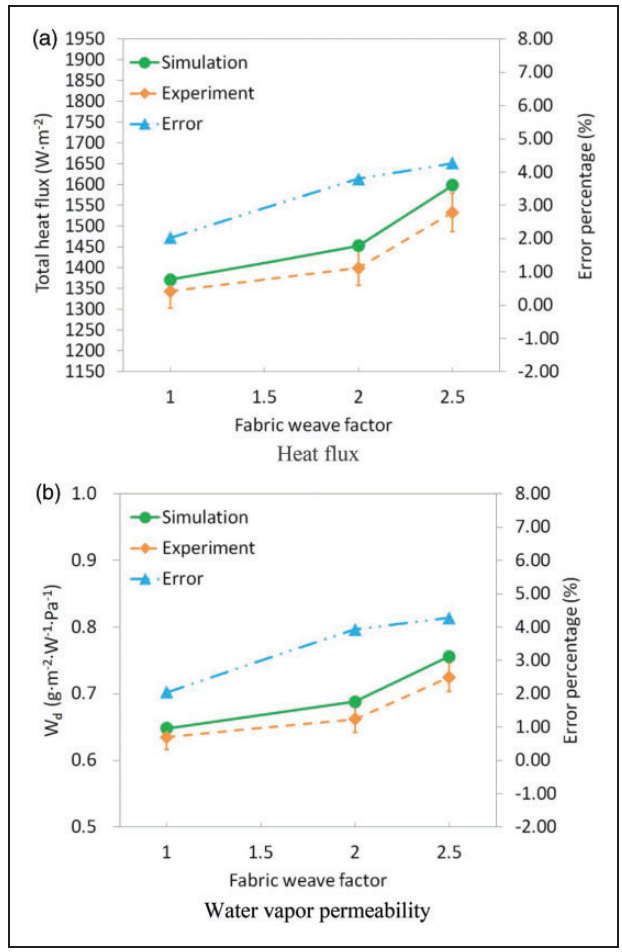

Figure 7. Comparison of simulated and experimental values: (a) heat flux and (b) water vapor permeability.

Discussion and prediction

Heat and moisture transfer under different fabric weave pattern

To investigate changes in heat and water vapor over time in plain, 2/2 twill, and 4/1 twill weave fabric models during heat and moisture transfer processes, each quantity was solved using a transient solver in the model between 0–2 s. The step size of the transient solver was set to 0.01 s. The alterations in heat conduction fluxes for all three fabric models can be seen in Figure 8. Due to the absence of a temperature difference between the upper and lower surfaces of the model, the heat conduction fluxes are minimal, measuring only 10^{-7} W/m^2. Figure 9 illustrates the fluctuation in convective heat flux among the three fabric models. The heat convection flux in the plain weave fabric increases rapidly to 1305.2 W/m^2 and then rises gradually to reach 1371.2 W/m^2, while maintaining a dynamic equilibrium. Similarly, the heat convection flux in the 2/2 twill weave fabric shows a sharp increase to 1398.4 W/m^2 followed by a slow increase to 1453.1 W/m^2 while remaining in dynamic equilibrium. The heat convection flux in the 4/1 twill weave fabric rises sharply to 1480.3 W/m^2 and then gradually increases to 1597.9 W/m^2 while maintaining dynamic equilibrium. Figure 10(a)–(c) displays the heat convection flux at different locations within the three fabric models. It is noticeable that the 4/1 twill weave fabric

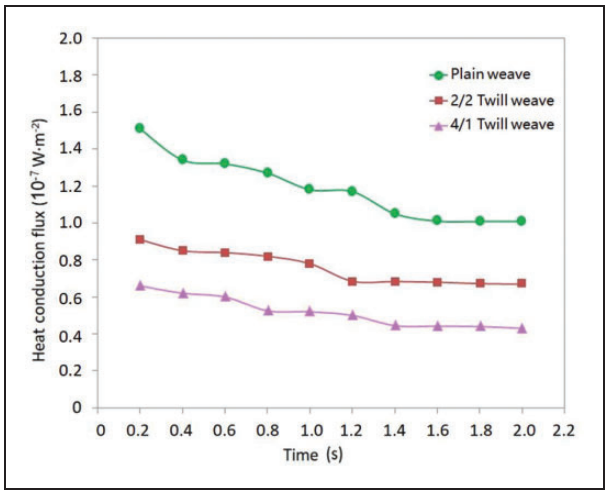

Figure 8. Heat conduction flux with time in the fabric model.

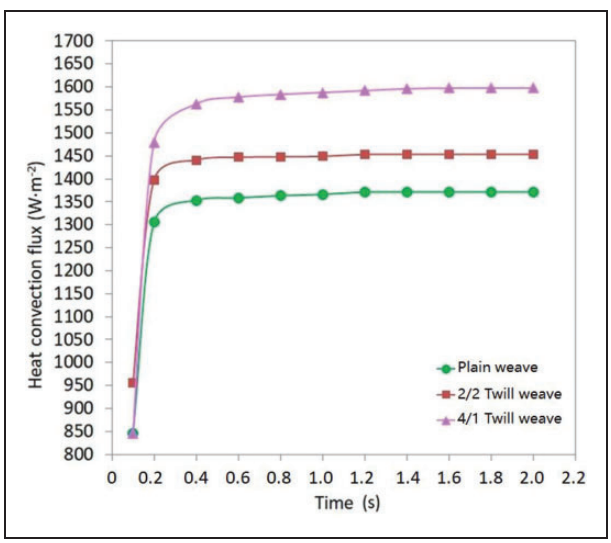

Figure 9. Heat convection flux with time in the fabric model.

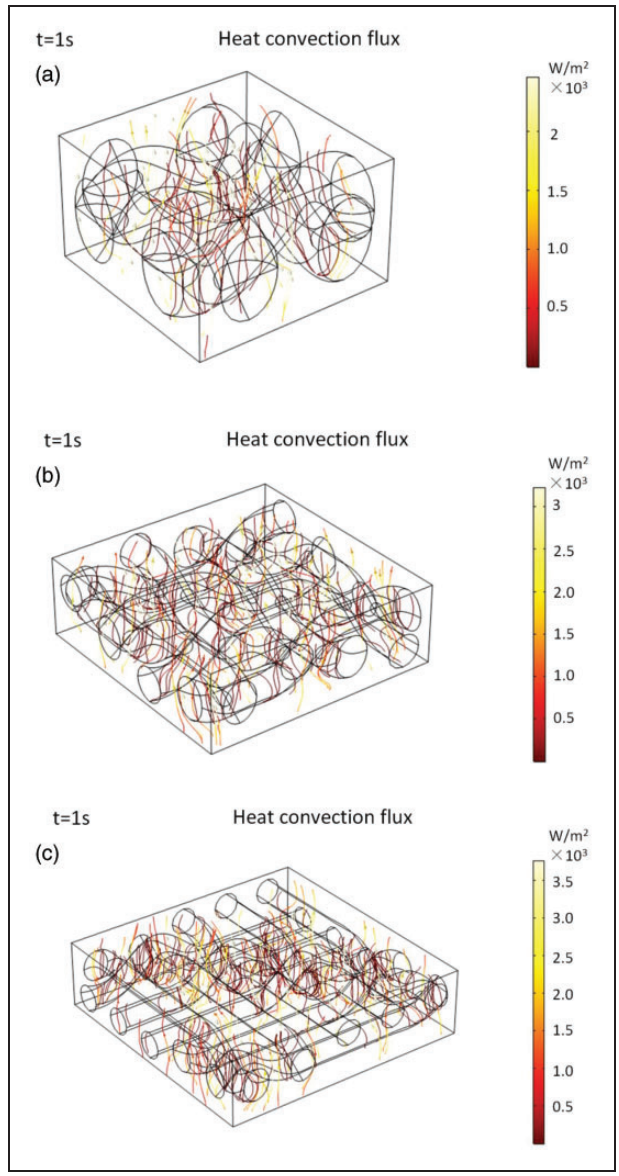

Figure 10. Heat convection flux at different locations in the fabric model: (a) plain weave fabric, (b) 2/2 twill weave fabric, and (c) 4/1 twill weave fabric.

exhibits the highest heat flux, followed by the 2/2 twill weave fabric, whereas the plain weave fabric displays the weakest heat flux.

During the process of water vapor transfer, concentration differences occur between the two sides of the fabric, which results in the diffusion of water vapor from the higher to the lower concentration area. In this model, the water vapor flows from the lower surface to the upper surface of the fabrics. Taking plain weave fabric as an example, Figure 11(a)–(d) shows the water vapor concentration on the lower surface of the fabric during the water vapor's diffusion and flow from bottom to top at 0 s, 0.5 s, 1 s, and 2 s, respectively.

Figure 12(a)–(c) displays the water vapor concentration on the lower surface of plain, 2/2 twill, and 4/1 twill weave fabrics at 0.5 s. Figure 13(a)–(c) illustrates the vertical distribution of water content of the three fabrics in a cross-sectional view at 1 s. It can be seen from the figures that, due to the different water vapor content on both sides of the fabric, the water molecules move from the lower surface towards the upper surface of the fabric. The degree of water vapor diffusion increases from the plain weave fabric to 2×2 twill to the 4/1 twill weave fabrics. Among the three fabrics, the plain weave fabric's interweaving is the most uniform geometrically (number of yarns per float is 1, known also as float length or weave factor, in other words there is a weave intersection per yarn within the repeat), resulting in the most uniform diffusion.

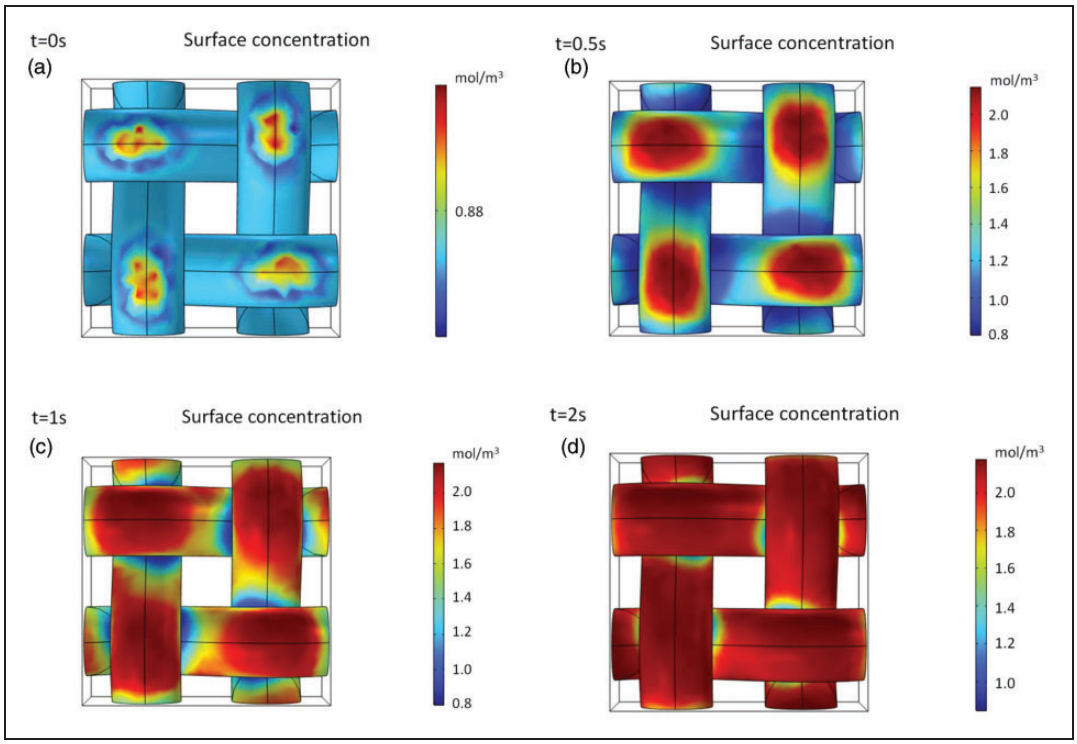

Figure 11. Water vapor diffusion with time on the lower surface of plain weave fabrics: (a) $t = 0$ s, (b) $t = 0.5$ s, (c) $t = 1$ s, and (d) $t = 2$ s.

The other two weaves are formed from weave intersections and floats. The weave factor for the 2/2 twill and 4/1 twill weaves are 2 and 2.5, respectively. The 2/2 twill weave exhibits the next level of uniformity following the plain weave as this particular weave is a 2/2 balanced twill where the warp and weft yarns share both sides of fabric equally. On the other hand, the 4/1 twill is unbalanced where one top side of the fabric is dominated by the warp yarns and the bottom side is dominated by the weft yarns. Moreover, the float length is unequal with float lengths of one and four.

Figure 14(a)–(c) depicts the water content distribution in the middle cross section of plain, 2/2 twill and 4/1 twill weave fabrics at $t = 1$ s. It is evident that the water vapor concentration in the middle cross section of the three fabrics increases as the weave factor increases. Figure 16(a)–(c) displays flow line diagrams of the water vapor concentration in the three fabrics at $t = 1$ s at different positions. Figure 15(a)–(c) illustrates the vertically downward water vapor concentration distribution over all three fabric models. Figure 17(a)–(c) exhibits the vertical downward fluid velocity distribution of the three fabrics at $t = 1$ s. From Figures 14–16, it is evident that water vapor passes through the fabric modeling process, with a preference for passing through the fabric from the central void and the void in the yarns. As seen in Figure 17, it is evident that there is a noticeable increase in fluid flow velocity from the bottom to the top of the fabric, which is due to the lower humidity of the upper surface compared to the lower surface. In addition, the central portion of the fabric experiences more profound fluid flow velocity compared to the two sides. The findings indicate that the variation in pore distribution resulting from the difference in fabric weave structure impacts the moisture transport of the material. The 4/1 twill weave fabric exhibits superior performance compared to 2/2 twill weave fabric, with plain weave fabric performing the worst.

From the simulation results and analysis above, it is evident that the heat convection flux and water vapor permeability in the plain, 2/2 twill, and 4/1 twill weave fabric models increase with the weave factor from 1 to 2.5. This suggests that the heat and moisture transfer performance of fabrics is influenced by the fabric weave pattern for a given yarn and fabric cover. Different fabric weave patterns result in fabrics with distinct pore distribution and interweaving characteristics. Among plain, 2/2 twill, and 4/1 twill weave fabric, plain weave contains the highest number of interweaving and the least amount of static air. The 4/1 twill weave fabric contains less interweaving, a longer average floating length, and a more concentrated pore distribution, leading to the improvement of heat and moisture transfer performance.

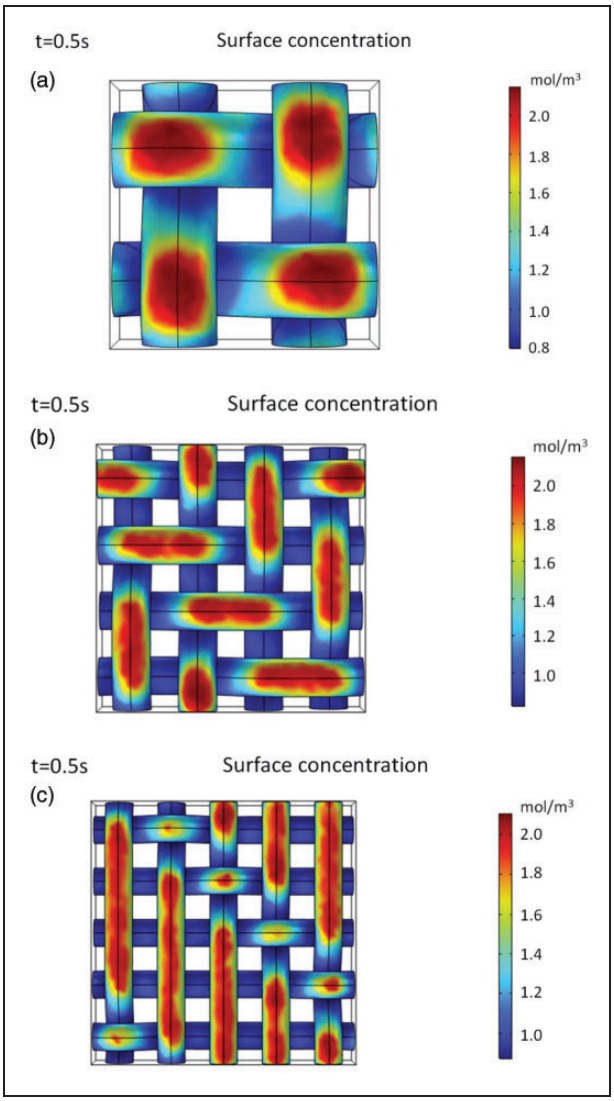

Figure 12. Water vapor concentration on the lower surface of the fabric model at $t = 0.5$ s: (a) plain weave fabric, (b) 2/2 twill weave fabric, and (c) 4/1 twill weave fabric.

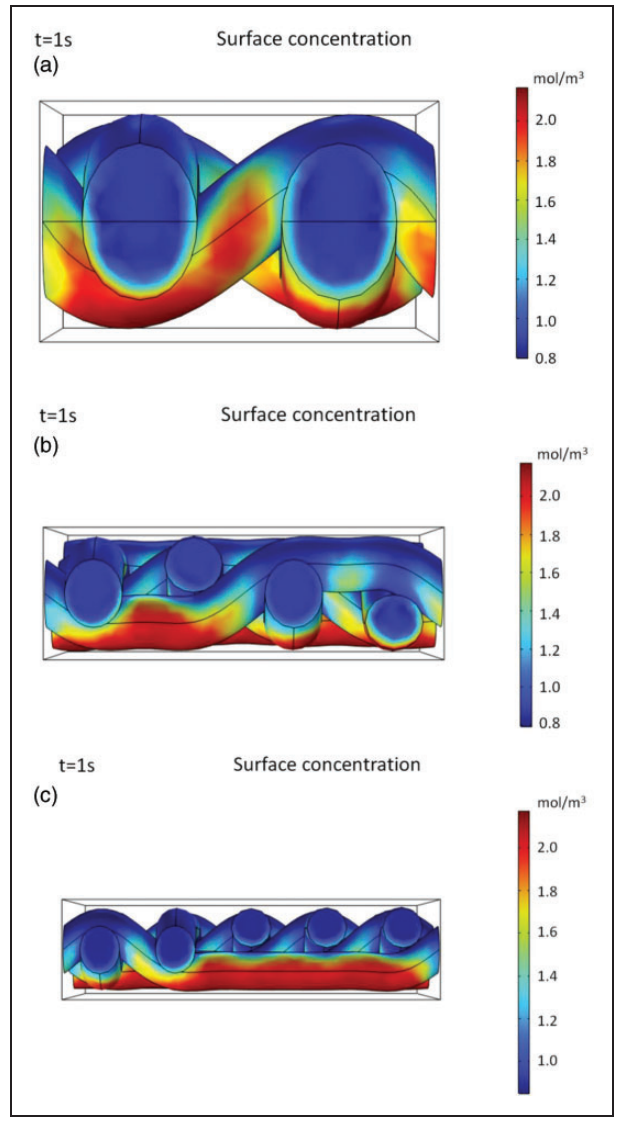

Figure 13. Vertical downward water vapor concentration for the fabric model: (a) plain weave fabric, (b) 2/2 twill weave fabric, and (c) 4/1 twill weave fabric.

Heat and moisture transfer under different fabric cover

In the preceding section, the effects of different fabric structures on fabric moisture transfer were explored. Here, the above model is employed to predict and analyze the alterations in heat and water vapor transfer for hypothetical woven fabrics with different covers.

Fabric cover, which is a measure of fabric degree of tightness,[31] is defined as the ratio of the projected yarn area to the total fabric area.[32] The fabric cover for square fabric ϕ can be expressed in terms of yarn spacing and diameter:

$$\phi = \frac{2L \cdot D - D^2}{L^2} \tag{17}$$

where L denotes yarn spacing, and D denotes yarn diameter.

Five models of plain weave fabrics with different covers (85.1%, 82.1%, 78.1%, 75%, and 72.1%) were constructed by varying the yarn spacing (L = 1.63D, 1.732D, 1.88D, 2D, and 2.12D, respectively) The yarn diameter D was kept constant at 0.34 mm. Here, thread densities (per 10 cm) of different hypothetical fabrics could be calculated respectively (180.4, 169.8, 156.4, 147.1, and 138.7). Figure 18(a)–(e) shows a visual presentation of the five hypothetical fabrics.

The physical field setup was executed using the method in the previous section "Construction of the FE model". The transient solver was used in the FE COMSOL software to solve for each quantity of the

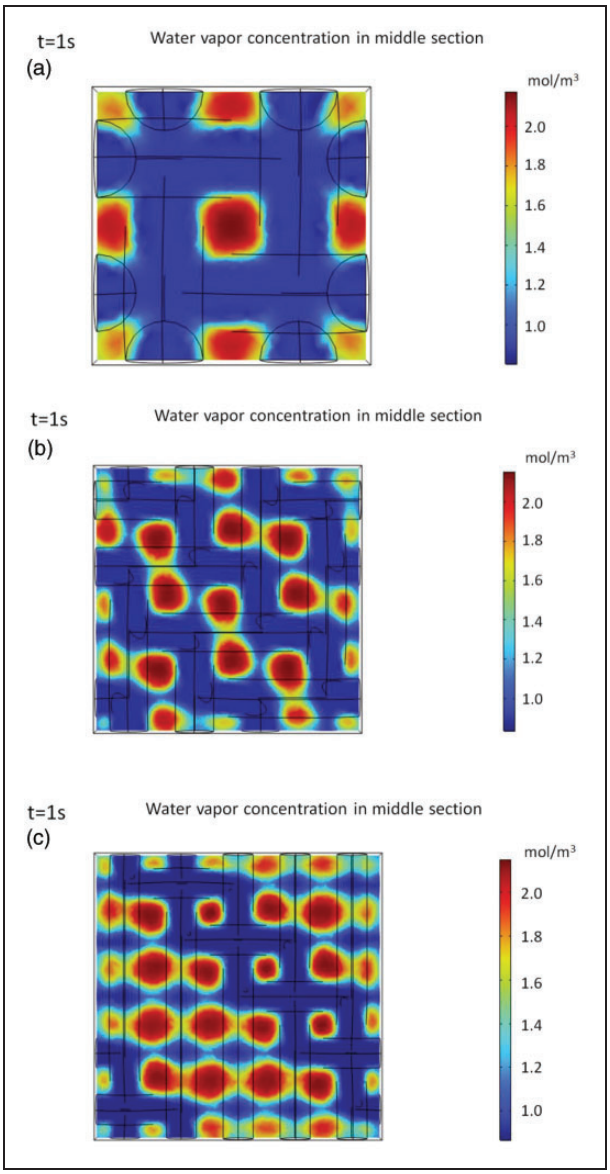

Figure 14. Water vapor concentration in the middle section of the fabric model: (a) plain weave fabric, (b) 2/2 twill weave fabric, and (c) 4/1 twill weave fabric.

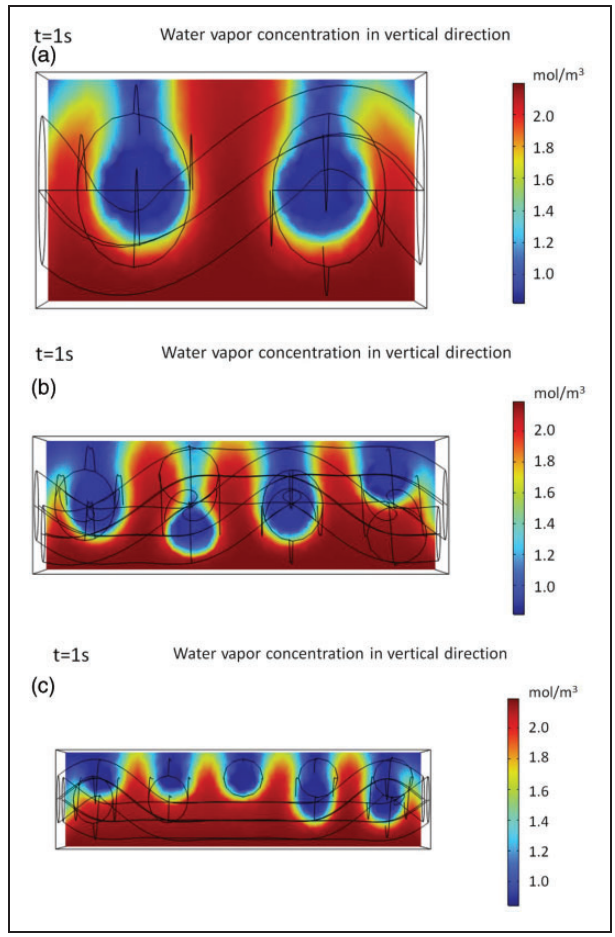

Figure 15. Water vapor concentration in the vertically downward section of the fabric model: (a) plain weave fabric, (b) 2/2 twill weave fabric, and (c) 4/1 twill weave fabric.

model in the 0–2 s time range, with a step size of 0.01 s. Figure 19(a)–(c) exhibits the heat fluxes in the fabric models with cover of 85.1%, 78.1%, and 72.1% at 0.5 s. Figure 20(a)–(c) displays the downward vertical concentration of water vapor within fabric models with cover of 85.1%, 78.1%, and 72.1% at 0.5 s. Observing this sequence indicates that the fabric models with lower cover demonstrate an increased capacity for heat and water vapor transfer.

The heat flux variations of five plain weave fabric models with different covers are exhibited in Figure 21, while Figure 22 shows the alterations in moisture resistance and water vapor permeability, converted from heat flux. It is evident that as the fabric cover drops gradually within the range 85.1%–78.1%, the heat flux rises notably. Concomitantly, moisture resistance declines steeply while water vapor permeability rises steeply. As the fabric cover further decreases and falls below approximately 75%, the heat flux no longer registers a significant increase and changes occur at a slower pace until reaching a stabilized state.

Based on the simulation results presented above, it is evident for a given fabric weave pattern, decreasing the fabric cover from 85.1% to 72.1%, the heat flux in the fabric model increases. Furthermore, the water vapor transmission performance is also enhanced. These findings suggest that the heat and moisture transmission performance is influenced by the fabric cover. As the fabric cover decreases from 85.1% to 78.1%, there is a marked change in the fabric's heat and moisture transfer performance. However, once the cover falls below approximately 75%, there is no longer a significant change. This suggests that within a certain range of cover, the fabric's heat and moisture transfer

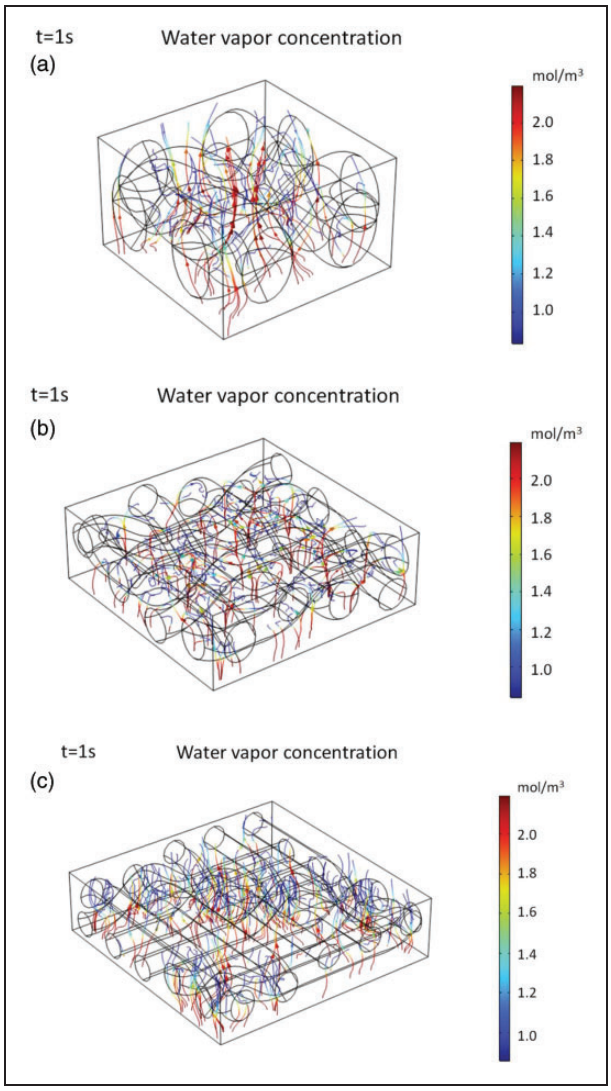

Figure 16. Water vapor concentration at different locations in the fabric model: (a) plain weave fabric, (b) 2/2 twill weave fabric, and (c) 4/1 twill weave fabric.

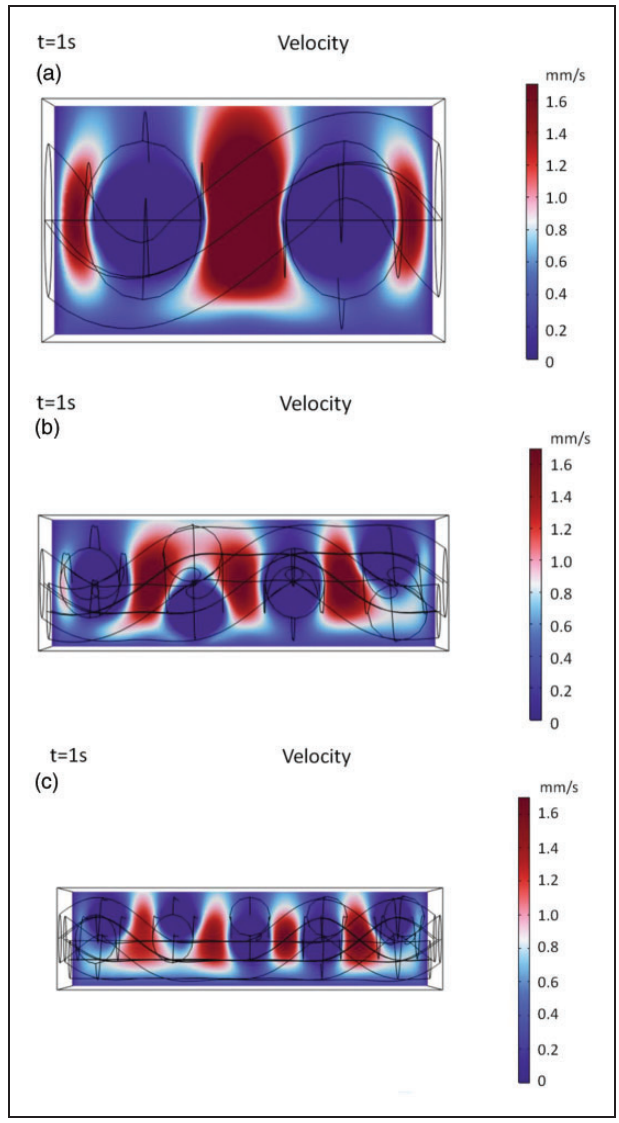

Figure 17. Velocity in the vertical section of the fabric model: (a) plain weave fabric, (b) 2/2 twill weave fabric, and (c) 4/1 twill weave fabric.

performance is influenced by pore size and distribution. However, once the cover decreases below a certain level, the enlarged pores on both sides of the fabric allow for easier air exchange and moisture transfer through convection in the micro-porous area. The performance of the fabric is no longer significantly impacted by the size and distribution of the pores.

Summary and conclusions

For the critical impact on consumer comfort, both heat and moisture transfer have been researched since the last century. A literature review disclosed that researchers did not consider fiber properties and cannot predict today's textiles fabrics that are made from wide range of natural, manufactured, and synthetic fibers and their blends. While the last few decades have witnessed a good number of publications in the field, researchers dealt with 1D or 2D aspects and did not consider the 3D nature of textiles, they did not make use of the power of FE methods and their software, and did not focus on heat or moisture transfer. More recent studies made use of FE models that resulted in better simulation; however, these models did not consider the effects of different fabric structures (such as weave structure, yarn density, yarn spacing, yarn diameter, fabric cover, and fabric degree of tightness) on both heat and moisture transfer of the fabric. To address the lack of previous work, in this article, we considered the influence of fabric cover of plain woven fabrics and weave structure for a given fabric cover on the heat and moisture

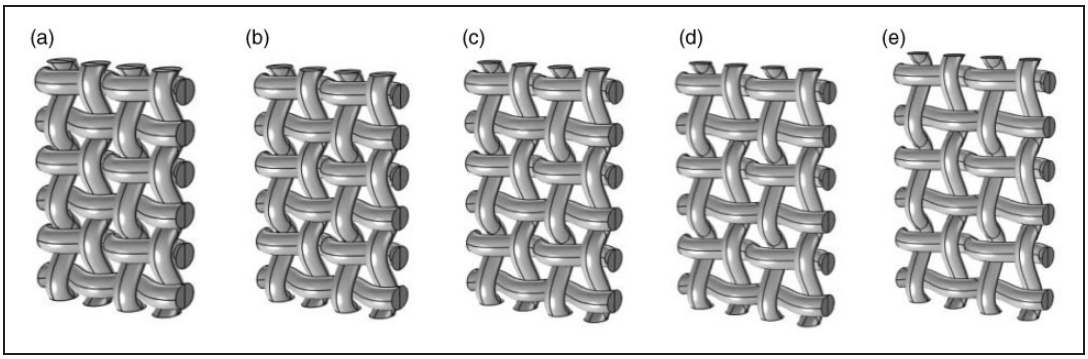

Figure 18. Plain weave fabric model with different covers: (a) 85.1% cover, (b) 82.1% cover, (c) 78.1% cover, (d) 75% cover, and (e) 72.1% cover.

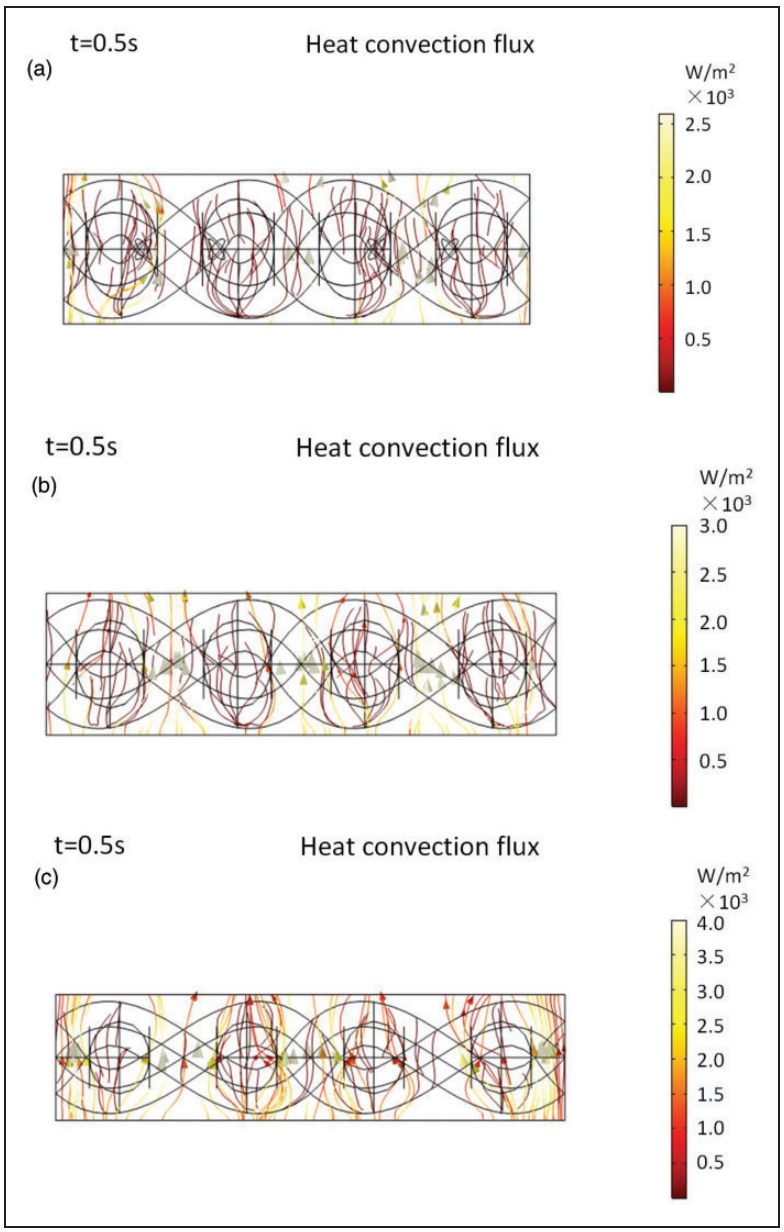

Figure 19. Heat fluxes of plain weave fabrics with different covers at 0.5 s: (a) 85.1% cover, (b) 78.1% cover, and (c) 72.1% cover.

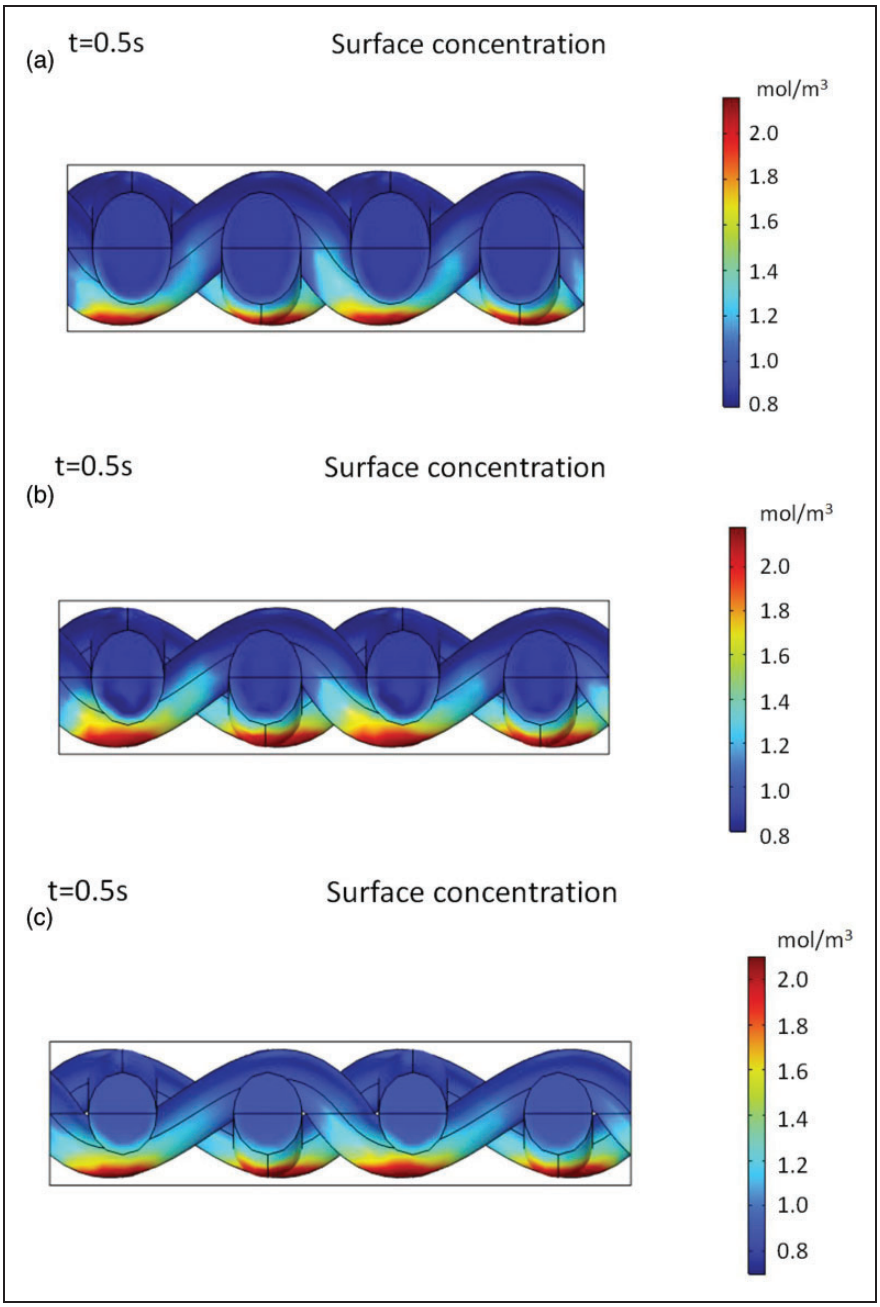

Figure 20. Water vapor concentration plain weave fabrics with different covers at 0.5 s: (a) 85.1% cover, (b) 78.1% cover, and (c) 72.1% cover.

transfer in woven fabrics using FE 3D simulation and experimental validation.

To investigate the influence of fabric structural parameters on the heat and moisture transfer performance of fabrics, coordinates of each control point corresponding to the yarns were obtained according to the interweaving characteristics of equal support surface fabrics (square fabrics). Interpolation curves were constructed using the method of interpolating functions, and the minimum fabric organization system model of plain, 2/2 twill, and 4/1 twill weave patterns containing stationary air was constructed using sweeping and arraying tools. The fabric model was analyzed sequentially using the FE analysis COMSOL software after setting parameters and boundary conditions, in accordance with the experimental principle of the thermal and moisture resistance tester to examine how fabric structural parameters affected the transfer of heat and water vapor. The changes in the diffusion of water vapor over time

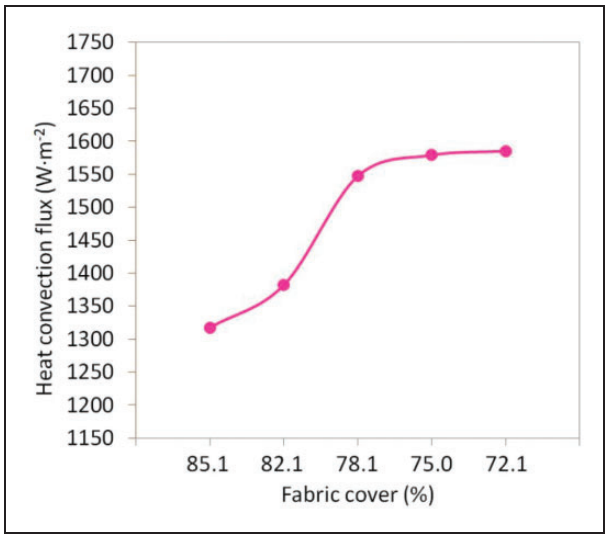

Figure 21. Heat flux of different fabrics' cover.

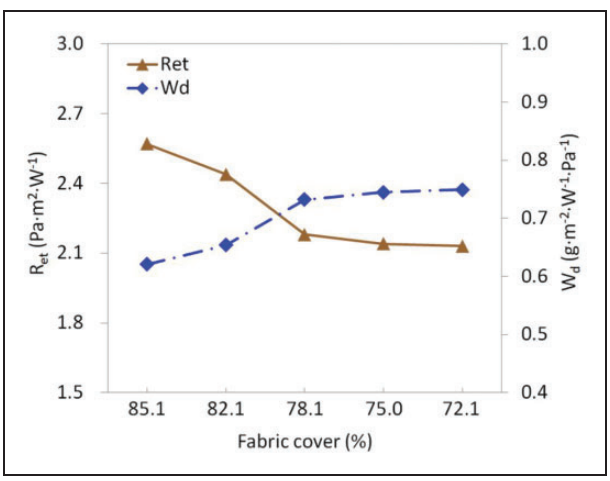

Figure 22. Moisture resistance and water vapor permeability of fabrics of different covers.

were identified through the graph of water vapor concentration distribution, while the water vapor concentration streamline diagram of the fabric model enabled visibility of the distribution of vapor concentration in different locations within the model. Both the moisture content distribution and fluid velocity distribution cross-sectional graphs demonstrate that the water vapor transfers preferentially through the pores while the flow rate in the middle of the fabric is larger. The changes of heat flux with time in the fabric models were analyzed through the conduction and convection heat flux with time graphs.

Based on the simulation results, a horizontal comparison was conducted on the heat and moisture transfer properties of plain, 2/2 twill, and 4/1 twill weave fabrics. In addition, the impact of varying fabric weave pattern on heat and water vapor transfer was examined. The findings of the simulation indicate that 4/1 twill weave fabric exhibits the highest level of heat and moisture transfer performance. The 2/2 twill weave fabric demonstrates the second-highest results, whereas plain weave fabric exhibits the least heat and moisture transfer properties. Our study indicates that the fabric weave structure has a significant impact on the heat and moisture transfer of fabrics. Woven fabrics with a high concentration of pores, longer float length, and more static air content have higher heat and moisture transfer performance.

By adjusting the control point coordinates, five samples of plain weave fabrics with varying levels of cover were created, and a longitudinal analysis was conducted to compare the heat and moisture transfer performance of these fabrics. The simulation results indicate that, when the fabric cover is within the range of 72.1%–85.1%, a gradual decrease in cover between 85.1% and 78% leads to significant changes in the fabric's heat and moisture transfer performance, but there is no longer a significant change when it is lower than about 75%. This change suggests that the heat and moisture transfer performance of the fabric is influenced by the size and distribution of the pores within a specific range of cover, the lower the cover the better the heat and moisture transfer properties of the fabric but, once below a certain cover, the enlarged pores on both sides of the fabric allow for easier air exchange and moisture transfer through convection in the micro-porous area. The performance of the fabric is no longer significantly affected by the pore size and distribution.

After verifying the 3D simulation's efficacy in this study, it became apparent that the difference between the heat flux in the process of water vapor transfer in the simulation and the experimental value is only within 5%. This finding indicates that the 3D simulation technique can realistically demonstrate and anticipate the impact of fabric weave pattern represented by weave factor (tightness) for a given fabric cover on the fabric's heat and moisture transfer performance. Consequently, this method can be used to examine the heat and water vapor distribution inside the fabric and provide a theoretical framework for the optimal design of the fabric's heat and moisture comfort.

In this article, our research is focused on polyester square fabrics (equal-supported fabrics) of different weave structures at a fixed fabric cover. We also demonstrated the 3D simulation for hypothetical square plain weave fabrics of different covers. Realizing the fact that numerous woven fabrics are constructed with unequal warp and weft geometrical parameters, known as non-square fabrics (warp-supported and

weft-supported fabrics) and different fiber types, our current and future research will address these fabrics utilizing the foundation work we established in this study.

Declaration of conflicting interests

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